Time to Shop Until Inflation Drops: The Effect of Inflation on Household Shopping Effort*

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Abstract

I study how households adapt their shopping behavior when inflation rises. Using transaction-level panel data from 2020–2023, I construct household-specific inflation exposure and estimate its causal effects on shopping intensity and bargain-hunting. Higher exposure leads to more trips, greater retailer diversification, and increased reliance on sales and coupons within trips. These adjustments entail real resource costs; using time-use data, I approximate the additional time households devote to price search and deal-finding. My findings suggest that households during the recent inflation episode absorbed part of the price shock by reallocating effort toward shopping.

JEL Classification: E31, D12, D83, E21, L81, C23.

Keywords: Inflation; Consumer search; Shopping intensity; Bargain hunting.

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1 Introduction

Inflation in the United States reached its highest level in four decades in June 2022. In the same year, consumer sentiment plummeted, and almost 50 percent of the Michigan Survey respondents cited higher prices as a reason for their deteriorating personal finances. Though the rate of these price increases has slowed considerably over the last few years, the year-over-year percent change in the headline Consumer Price Index was still above the Fed's target of 2% as of August 2025. The question arises of how this historic and arguably persistent run-up in prices, coupled with a considerable decline in consumer sentiment, translates to real effects on household behavior and consumption.

In this paper, I study how households adapt their shopping behavior in response to realized inflation. While economists typically measure inflation using aggregate indices such as the CPI, households experience inflation through the evolving costs of the goods they purchase each week. It is not the "representative basket" of the CPI, but the prices of the specific items in their own baskets, that shape households' consumption decisions. These decisions often occur on the margin: whether to switch to generic products, visit another store to take advantage of a sale, or spend additional time searching for lower prices. Such behaviors resemble classic "shoe-leather costs" of inflation, as households expend limited resources like time to reduce expenditure.

Recent survey evidence suggests that bargain-hunting is one of the primary ways households cope with high prices. Data from the Federal Reserve's Survey of Household Economics and Decision-making and the Census Bureau's Household Pulse Survey show that a majority of households in 2022 reported shopping at cheaper stores, looking for sales, using coupons, or switching to lower-priced products in response to inflation. This is consistent with standard sticky-price models of inflation with consumer search: when there is inflation and there is a cost for sellers to update prices, households can exploit the subsequent variation in prices by searching around at different retailers for the lowest prices for identical goods (Benabou, 1988).

While aggregate inflation provides a useful summary of economy-wide price trends, households face substantial heterogeneity in their actual inflation rates (Kaplan and Schulhofer-Wohl, 2017). This heterogeneity arises from two sources: households pay different prices for identical goods, and they purchase different bundles of goods, creating differential exposure to price changes across categories. Using transaction-level purchase data from the NielsenIQ Homescan Consumer Panel, I quantify how inflation increased shopping intensity and bargain-hunting behavior from 2020 to 2023. For my empirical strategy, I employ a Bartik-style shift-share instrument (SSIV) that combines temporal variation in national inflation at the CPI-stratum level with cross-sectional variation in predetermined household consumption patterns. The SSIV captures variation in potential inflation exposure that is: (1) tied to a household's consumption composition, and (2) realized only through national price shifts that households could not anticipate when forming their 2019 consumption shares. The instrument's relevance stems from differential volatility across product categories, in that some goods experienced more dramatic price swings than others during the most recent inflationary period. I see whether greater exposure to price swings caused households to partake in more bargain-hunting, and I provide a back-of-the-envelope calculation that suggests how costly these behaviors are in terms of time.

My results show that a 1 percentage point increase in household-level inflation causes households to undertake, on average, 1.95 additional shopping trips per month (approximately 16 percent relative to the mean) and to visit 0.59 more distinct retailers (approximately 11 percent). The extra trips are strategically allocated: roughly 35 percent go to discount stores, dollar stores, and warehouse clubs, retailer types that offer systematically lower prices. On the intensive margin of search, households increase the share of trips using coupons by 0.81 percentage points (approximately 6 percent) and the share of trips with at least one item on sale by 0.91 percentage points (approximately 3 percent). Households also show evidence of stocking-up behavior, with items per trip rising by 1.00 (approximately 8 percent) and generic share increasing modestly by 0.29 percentage points (approximately 2 percent). These magnitudes are substantially larger than OLS estimates and often opposite in sign, indicating that simultaneity and measurement error bias conceal the true behavioral responses of households.

These behavioral adjustments represent a real cost of inflation beyond the direct erosion of purchasing power. While the search effort households exert is individually rational and helps them maintain consumption in the face of rising prices, from a social perspective this effort is largely unproductive. In a frictionless benchmark, households would not need to devote time and attention to comparing prices across stores, timing trips to sales or promos, or trying to find better deals at different types of retailers. The fact that inflation induces such effort reflects underlying price rigidities and dispersion that create privately profitable but socially wasteful search. I provide a back-of-the-envelope calculation that translates the increase in trip frequency into time costs using data from the American Time Use Survey, suggesting that even moderate inflation episodes impose non-trivial time burdens on households. This shoe-leather cost should be incorporated into welfare assessments of inflation, particularly when evaluating the trade-offs between price stability and other macroeconomic objectives.

I make contributions to several strands of literature. First, my study contributes directly to the literature on consumer search, and in particular, bargain-hunting behavior. Lee et al. (2021) show using granular expenditure data that households bargain-hunt for the goods they prefer and pay lower prices, as a test of rational consumer search. Related work also documents how search and shopping strategies shape price dispersion and household expenditure patterns (e.g., Cox et al., 2020; Cavallo and Kryvtsov, 2024; Petrosky-Nadeau et al., 2016; Lee, 2024; Kaplan, 2017; Gauri et al., 2008; Griffith et al., 2009; Janssen and Kasinger, 2025; Pytka, 2024).

My paper also contributes to research that analyzes household consumption responses to inflation expectations and realizations. Ryngaert (2022) uses the New York Fed's Survey of Consumer Expectations to show that households anticipating inflation "disasters" alter their purchasing decisions even before experiencing higher prices. My findings complement this work by documenting behavioral changes in response to realized inflation of disaster-magnitude proportions (see also Schnorpfeil, Weber, and Hackethal, Schnorpfeil et al.; D'Acunto et al., 2021; McConnell et al., 2010).

This paper also contributes to the inflation measurement literature. Kaplan and Schulhofer-Wohl (2017) apply traditional price indices to household expenditures and find large heterogeneity in inflation rates across households, mostly due to variation in the prices paid for identical goods. Jaravel and O'Connell (2020) highlight how shopping changes affect inflation measurement, while Pathak Chalise (2025) focus on how shopping strategies reduce observed inflation.

My findings also connect to research on home production as a margin of adjustment during deteriorating macroeconomic conditions. Following canonical models such as Becker (1965), studies show that households increase time allocated to activities like cooking, cleaning, and shopping during recessions (Aguiar and Hurst, 2007; Aguiar et al., 2013; Nevo and Wong, 2019; Cacciatore et al., 2023). Other work emphasizes substitution toward lower-quality goods, changes in consumption bundles, or increased housework in response to various kinds of shocks (Jaimovich et al., 2019; Lester, 2014; Gutiérrez-Daza, Gutiérrez-Daza; Guillochon, 2024; Fang et al., 2020; Rupert et al., 1995; Been et al., 2020). My paper provides new evidence on budget-tightening shopping strategies during the recent inflationary episode.

I also contribute to the large literature that measures the welfare costs of inflation and the role of search. Sara-Zaror (2024) focuses on how inflation affects the relative prices households pay through returns to search, with welfare implications stemming from both the dispersion households exploit and the resources they expend in doing so. Building on hypotheses from models of consumer search and price dispersion (Benabou, 1988; Diamond, 1993; Head and Kumar, 2005; Wang, 2016), my paper emphasizes the behavioral responses of households rather than equilibrium price-setting, showing empirically that inflation prompts costly search and adjustment at scale. This evidence can help discipline models of inflation by quantifying the time and resource costs borne by households.

This paper further relates to work linking inflation, search, and inequality, including various studies concerning heterogeneity in search effort, geospatial differences in inflation, and inequality in product variety or innovation (Boerma and Karabarbounis, 2021; Coibion et al., 2021; Nord, 2022; Navarrete and Kim, 2024; Bui, Bui; Jaravel, 2019; Chen et al., 2024). My results complement this literature by showing how realized inflation exposure induces differential adjustments across households, with potential implications for inequality and welfare.

Lastly, my paper speaks to the long-standing but recently revived literature that asks: why do people dislike inflation? Shiller (1997) posed the question, and more recent work finds that weak wage pass-through and diminished purchasing power force households to undertake costly adjustments (Hajdini et al., 2025; Stantcheva, 2024). I provide causal evidence that greater inflation exposure prompts such costly behavioral changes, consistent with both survey evidence and observed household spending decisions.

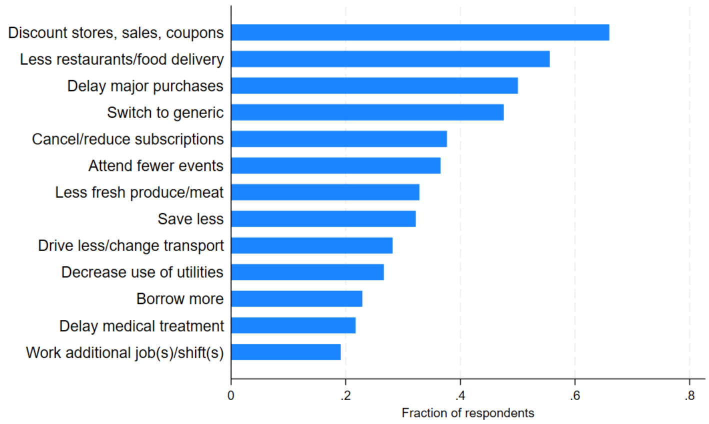
The rest of the paper proceeds as follows. Section 2 provides additional analysis of self-reported survey responses to identify the range of behaviors households adopted in response to inflation. Section 3 describes a simple consumer choice framework to fix ideas about how households can exert shopping effort as a way to lower their paid prices. Section 4 describes the datasets I use to construct household-specific inflation rates and shopping outcomes for my main empirical analysis. Section 5 outlines my empirical methodology, while Section 6 presents baseline results on how inflation exposure affects shopping effort and bargain-hunting behavior. Section 7 concludes.

2 Survey Evidence on Household Responses to Inflation

In this section, I use survey microdata from the recent inflationary period to identify what behaviors households employed when facing higher prices. By analyzing self-reported coping behaviors, I provide suggestive evidence regarding along what dimensions inflation is potentially costly for households based on how they adjust.

In 2021 and 2022, surveys began to ask households about their attitudes and actions regarding the increase in prices, including the Household Pulse Survey, conducted by the Census Bureau since the onset of the COVID-19 pandemic. The survey fielded a subset of questions related to inflationary pressures and households' resulting behavior. They are asked the question: *"What changes, if any, have you made to cope with the increase in prices? (Select all that apply)."* Figure 1 shows each coping strategy the household could select, where shopping at discount stores, buying items on sale, and using coupons was the most commonly selected response.

Figure 1. Household responses to higher prices - Pulse.



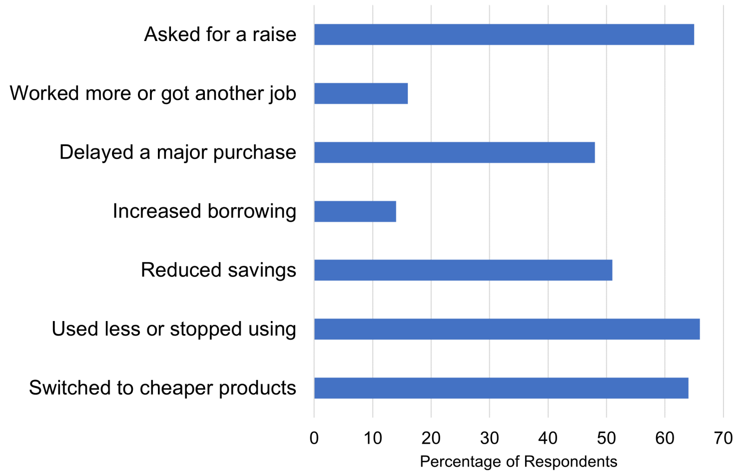
Source: Census. Sample period: September 2022 to June 2023.

Similarly, the Survey of Household Economics and Decision-making (SHED) from the Federal Reserve Board of Governors asked about inflation coping mechanisms. The SHED is an annual, nationally representative survey, with the 2022 wave considering topics such as price changes' effects on households' budgets and shopping behavior, as well as their online shopping habits.[1] Figure 2 shows responses to the question: *"Did you take any of the following actions because of increases in prices over the past 12 months?"*. The first and third most commonly-chosen responses out of the SHED survey regarding coping behaviors for inflation were "used less or stopped using," and "switched to cheaper products," indicating that as a result of experiencing higher prices, households were cutting back (or out entirely) or substituting away from certain products that before they were purchasing as part of their consumption bundle. In other words, this evidence that households experienced an income effect over this period, decreasing their consumption due to their lowered purchasing power. Though I do not focus on substitution as a main outcome, I discuss how substitution can be measured using the price indexes I construct in this paper in Appendix B.

Data from the American Time Use Survey's 2022-2023 eating and health module provides additional context about household shopping preferences. When asked about online grocery shopping behavior, respondents revealed their primary reason for online shopping at all in the last month was because of time constraints (depicted in Figure 3), while their primary reason for avoiding online shopping was preferring to see and touch the items in the store.

[1]The survey for the 2022 data release was fielded in October 2022 after peak inflation numbers were realized, and so provides insight into the behavior of households who had already experienced peak price increases.

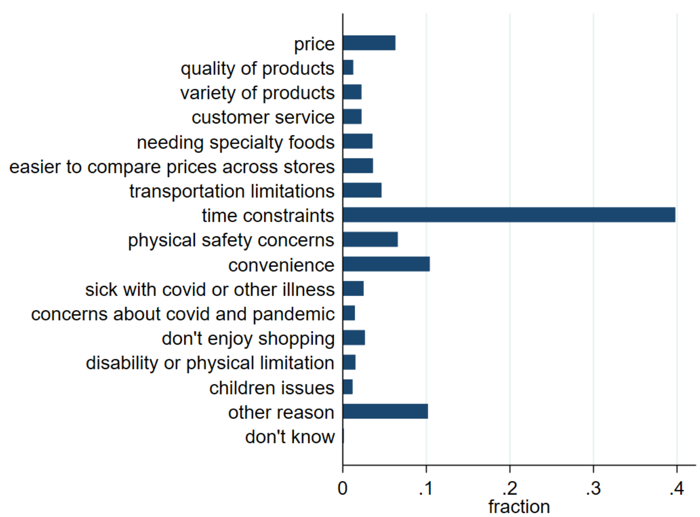
Figure 2. Household responses to higher prices - SHED.



Source: Board of Governors.

This evidence highlights the time cost dimension of shopping behavior for some households. I revisit the time use data in Section 6.2 to obtain a back-of-the-envelope cost of inflation in terms of shopping time.

Figure 3. ATUS: Main Reason for Grocery Shopping Online.



3 Inflation and Consumer Search

In this section, I motivate my empirical strategy with a simple model that illustrates how search behavior responds to inflation, and why OLS understates this relationship. The model captures the intuition that higher experienced inflation encourages households to search, but search in turn reduces their realized inflation.

3.1 Identification Problem

I begin with a brief description of the econometric problem. Let $s_{i,t}$ denote household i's search intensity in period t. Let $\pi_{i,t}^{\text{posted}}$ be the inflation rate the household faces before it decides how much to search (based on the posted prices it observes), and let $\pi_{i,t}^{\text{paid}}$ be the realized inflation computed from the prices the household actually pays (after search). I posit the following relationships between these objects:

$$s_{i,t} = \alpha_1 + \beta_1 \, \pi_{i,t}^{\text{posted}} + u_{1it}, \tag{1}$$

$$\pi_{i,t}^{\text{paid}} = \alpha_2 + \beta_2 \, s_{i,t} + \beta_3 \, \pi_{i,t}^{\text{posted}} + u_{2it}, \tag{2}$$

with $\beta_1 > 0$ (higher posted inflation raises search) and $\beta_2 < 0$ (search lowers paid inflation). The problem arises due to the fact that in the data, I do not observe $\pi_{i,t}^{\text{posted}}$ directly; what I actually observe is $\pi_{i,t}^{\text{paid}}$. In the reduced-form OLS regression of $s_{i,t}$ on $\pi_{i,t}^{\text{paid}}$,

$$s_{i,t} = \alpha_3 + \beta_4 \, \pi_{i,t}^{\text{paid}} + u_{3it}, \tag{3}$$

substituting in (2) for $\pi_{i,t}^{paid}$ makes the simultaneity across equations clear, as $\pi_{i,t}^{paid}$ is itself a function of $s_{i,t}$.

3.2 Simple Model

To formalize this intuition, I now lay out a deliberately stylized household decision problem. Consider a one-good environment. In period t the household sees a posted high price p_t^H and can, by exerting search effort $s_t \in [0, 1]$, raise the probability of paying a lower price $p_t^L < p_t^H$. Let the gap be $\epsilon_t \equiv p_t^H - p_t^L \geq 0$. I can write the expected price in a tractable way:

$$p_t(s_t) \;=\; (1 - s_t)p_t^H + s_t p_t^L \;=\; p_t^H - s_t \, \epsilon_t. \tag{4}$$

The household first observes (p_t^H, ϵ_t), chooses s_t, then pays $p_t(s_t)$. Households use their endowment y_t to consume c_t, so $c_t = y_t / p_t(s_t)$. I define the posted inflation rate as

$$\pi_t^{\text{posted}} \;\equiv\; \frac{p_t^H - p_{t-1}^H}{p_{t-1}^H},$$

and the paid inflation rate as

$$\pi_t^{\text{paid}} \;\equiv\; \frac{p_t(s_t) - p_{t-1}(s_{t-1})}{p_{t-1}(s_{t-1})}.$$

Let $u(c) = \log c$, and let there be a linear per-period search cost $k(s_t) = \kappa s_t$ with $\kappa > 0$. The household solves

$$\max_{s_t \in [0,1]} \; \log\!\left(\frac{y_t}{p_t(s_t)}\right) - \kappa s_t.$$

Using (4) and $p_t'(s_t) = -\epsilon_t$, the FOC is

$$\frac{\epsilon_t}{p_t(s_t)} \;=\; \kappa. \tag{5}$$

This yields an interior solution:

$$p_t(s_t) = \frac{\epsilon_t}{\kappa} \quad\Longrightarrow\quad s_t \;=\; \frac{p_t^H}{\epsilon_t} - \frac{1}{\kappa}. \tag{6}$$

If I assume that $\epsilon_t \approx \epsilon_{t-1} \equiv \epsilon$, I can write changes in search as

$$\Delta s_t \;\equiv\; s_t - s_{t-1} \;=\; \frac{p_t^H - p_{t-1}^H}{\epsilon_t} \;=\; \frac{p_{t-1}^H}{\epsilon_t}\,\pi_t^{\text{posted}}. \tag{7}$$

I want to note that the theory here pins down s_t as a function of π_t^{posted} (or p_t^H), not π_t^{paid}. In other words, Equation (7) shows that the marginal benefit of search depends on the posted price level p_t^H, which is observed before search. Again, this is because the household chooses s_t before it knows the realized paid price; hence the FOC conditions on posted prices (or posted inflation) and the distribution of discounts (ϵ_t).

Using (4), search maps back into paid inflation:

$$\begin{aligned}
\pi_t^{\text{paid}} &= \frac{p_t^H - s_t\epsilon_t - \left(p_{t-1}^H - s_{t-1}\epsilon_{t-1}\right)}{p_{t-1}^H - s_{t-1}\epsilon_{t-1}} \\
&= \underbrace{\frac{p_t^H - p_{t-1}^H}{p_{t-1}^H - s_{t-1}\epsilon_{t-1}}}_{\text{posted-inflation component}} \;-\; \underbrace{\frac{s_t\epsilon_t - s_{t-1}\epsilon_{t-1}}{p_{t-1}^H - s_{t-1}\epsilon_{t-1}}}_{\text{search component}}.
\end{aligned} \tag{8}$$

With the approximation $\epsilon_t \approx \epsilon_{t-1} \equiv \epsilon$ and assuming small s_{t-1}, I can express this in the following reduced-form way:

$$\pi_t^{\text{paid}} \;\approx\; \beta_1\,\pi_t^{\text{posted}} \;-\; \beta_2\,s_t \;+\; \beta_2\,s_{t-1}, \qquad \beta_1 \equiv \frac{p_{t-1}^H}{p_{t-1}^H - s_{t-1}\epsilon},\; \beta_2 \equiv \frac{\epsilon}{p_{t-1}^H - s_{t-1}\epsilon}. \tag{9}$$

Combining (7) with (9) yields the simultaneity in (1)–(2): posted inflation raises search, which then reduces paid inflation. Regressing $s_{i,t}$ on $\pi_{i,t}^{\text{paid}}$ thus combines the two effects, biasing β_4 in (3) toward zero (or negative). This is the core identification problem I address in Section 4, using an instrument that incorporates price movements that are correlated with the households' but that do not mechanically depend on household-level search intensity.

4 Data

In this section, I describe how I construct the individual inflation estimates and shopping outcomes from transaction-level purchase data and present some descriptive statistics both cross-sectionally and over time for these measures.

4.1 NielsenIQ Homescan Consumer Panel

My analysis relies primarily on household-level purchase data from NielsenIQ's Homescan Consumer Panel Survey. The panel tracks the shopping behavior of randomly recruited households who record all purchases by scanning product barcodes with a handheld device or mobile app. Households receive digital rewards for consistent participation, and must maintain regular scanning activity to remain in the panel. Each year there are roughly 60,000 respondents in the panel, and NielsenIQ retains about 80 percent of the active panel each year. NielsenIQ collects demographic information for households and updates this information on an annual basis.

For each shopping trip, households record the trip date and store name. For stores covered by NielsenIQ, the price of each scanned good is automatically set to the store's average price during the purchase week. For uncovered retailers, panelists manually enter prices. They also record the number of units purchased, whether they perceived the item to have been on a "deal," whether they used a coupon, and the total discount from that coupon.

NielsenIQ organizes purchases hierarchically from individual UPCs to broad departmental categories. The intermediate level consists of approximately 125 "product groups" (eggs, fresh meat, kitchen gadgets, pet care, etc.) that I use to match with CPI subcategory inflation rates for my shift-share instrument construction. I describe this CPI-stratum to NielsenIQ-product-group matching in greater detail in Appendix C. My analysis focuses primarily on non-durables, as this is mostly what is scanned by households in the NielsenIQ data.[2]

[2] In 2019, 54 percent of purchases in the full dataset were made at grocery stores, 28 percent at discount stores, 5 percent at warehouse clubs, 4 percent at dollar stores, 2 percent at drug stores, and 1 percent online shopping. The rest of the retailer types in my dataset account for less than 1 percent of the purchases.

These features of the data allow me to construct both household inflation rates and the shopping outcomes of interest for my main analysis. This dataset is uniquely well suited to the question of whether, and to what extent, inflation leads households to exert more shopping effort, as it provides exactly the combination of features needed: it follows the same households over time, records both prices and quantities for every scanned purchase, and includes rich demographic information. Together, these elements allow me measure directly how households adjust their shopping behaviors in response to inflationary pressures. While the data are self-reported and households may not scan every purchase, they nonetheless offer an opportunity to quantify the magnitude of these behavioral changes, assess heterogeneity across households, and trace how responses evolve over time.

4.2 Search Effort and Bargain-Hunting Measures

I construct a comprehensive set of outcome variables to capture household responses to inflation across multiple dimensions of shopping behavior. These measures fall into four conceptually distinct categories: search intensity on the extensive margin, search intensity on the intensive margin, retailer type substitution, and basket composition. Together, these measures provide a granular view of how households adjust shopping behavior in response to inflation.

Search intensity: Extensive margin. I begin my analysis by measuring how households expand search effort by increasing the frequency and scope of their shopping. The first outcome is the number of shopping trips per month, where a trip is defined at the store-date level using unique trip codes provided in the Consumer Panel. Purchases at different stores on the same day count as separate trips, as do purchases at the same store on different days. This measure captures one key margin of adjustment: households may increase trip frequency to better time purchases with temporary sales and promotions, a prediction that emerges naturally from search models where optimal search intensity rises with expected gains from finding lower prices.

The second extensive-margin outcome is the number of distinct retailers visited monthly, which I identify using unique store codes in the data. This retailer diversification measure captures cross-store price comparison shopping. When inflation raises price dispersion across retailers (for instance, because some stores adjust prices more slowly than others), the returns to visiting multiple outlets increase. A related outcome is trips per store, computed as the ratio of total monthly trips to distinct retailers visited. This measure helps distinguish between two forms of diversification: spreading a fixed number of trips across more stores (trips per store falls) versus adding new stores while maintaining trip frequency to existing ones (trips per store remaining stable or rising).

Search intensity: Intensive margin. Beyond expanding the number and scope of trips, households can intensify search within trips by more actively seeking and exploiting deals. I construct two trip-level measures of this intensive-margin bargain hunting. The first is an indicator for whether a household used any coupon during the trip, which I compute from coupon value fields in the purchase data. The second is an indicator for whether the trip included at least one item purchased on sale, identified using deal flags in the data. In my empirical analysis, I compute the share of a household's trips in a given month on which it used any coupon and the share on which it purchased any item on sale. These measures capture how thoroughly households exploit cost-saving opportunities conditional on shopping, complementing the extensive-margin search outcomes.

Retailer type. Households may respond to inflation not only by searching more but also by reallocating trips toward retailers that offer systematically lower prices. I construct trip counts by store type using retailer channel codes in the Consumer Panel, focusing on three categories where price savings are typically largest: discount retailers, dollar stores, and warehouse clubs. These type-specific trip measures capture strategic reallocation toward channels known for everyday low pricing, bulk discounts, or have a large amount of generic or private-label goods. In addition to absolute trip counts, I examine the share of total trips allocated to each type to distinguish pure trip expansion from substitution across channels.

Basket composition. Finally, I examine how inflation affects what households buy and how much they purchase per trip. The first outcome in this category is generic share, defined as the fraction of a household's monthly expenditure on store-brand or private-label products, identified using brand codes in the data. An increase in generic share indicates quality downgrading or brand substitution.

I also construct three measures of basket scale and scope. Items per trip is the monthly average number of distinct items purchased per trip, capturing whether households consolidate purchases into fewer, larger trips or break up shopping into smaller baskets. Items per pack measures the average pack size or unit count per purchased item, capturing bulk purchasing or package size substitution. Finally, the number of product groups purchased is the count of distinct NielsenIQ product groups (categories) that a household buys in a given month, capturing the breadth of a household's consumption basket. This last measure is particularly informative about the scope of a household's search: an increase in product groups suggests broader comparison shopping and more active category-level substitution, consistent with households spreading search effort across a wider range of goods.

4.3 Measuring Household-Level Inflation Rates

Understanding how inflation affects shopping behavior requires measuring the actual price changes that individual households experience, which may differ substantially from aggregate statistics. In this subsection, I describe how I construct these household-level inflation measures from the Consumer Panel. Two key methodological choices characterize my analysis. First, I use a Laspeyres price index, which holds the initial-period basket fixed rather than allowing quantities to adjust. Second, I measure inflation as year-over-year percent changes rather than month-over-month or cumulative changes. Both choices are motivated by the goal of isolating exogenous variation in price exposure from endogenous behavioral responses.

Following Kaplan and Schulhofer-Wohl (2017), I start with computing a Laspeyres index which holds the initial-period basket fixed. For each household i, I observe the quantity of each good j (identified by barcode) purchased in month t and its associated price $p_{ij,t}$ in both month t and month $t+12$. [3] The Laspeyres price index compares the cost of the period-t basket at $t+12$ prices to its cost at t prices. The household inflation rates I construct are based on matched barcodes only, i.e. a barcode would have to be present in both January 2021 and January 2022 to be used in the final calculation. I again follow Kaplan and Schulhofer-Wohl (2017) in requiring that a household have at least five matched barcodes in a given month in order to calculate their inflation rate. In Figure **??**, I compare the distributions of household inflation rates that are constructed using 5 matched barcodes and 25 matched barcodes.

$$P^L_{i,t,t+12} = \frac{\sum_j p_{ij,t+12}\, q_{ij,t}}{\sum_j p_{ij,t}\, q_{ij,t}}. \tag{10}$$

By construction, $P^L_{i,t,t} = 1$, so the associated inflation rate is simply the percent change in the index:

$$\pi^L_{i,t,t+12} = (P^L_{i,t,t+12} - 1) * 100. \tag{11}$$

I use the Laspeyres index because it holds the initial basket fixed, ensuring that my inflation measure is not mechanically influenced by any substitution that households may undertake in response to rising prices. In this way, the analysis isolates how inflation exposure affects search and deal-seeking behaviors, separately from the substitution channel. The Laspeyres index is also the natural analog to the Consumer Price Index, which similarly holds basket weights fixed between periodic updates. This comparability is important both for

[3] If a household buys the same barcode more than once in a month, I follow Kaplan and Schulhofer-Wohl (2017) and set the monthly price equal to the volume-weighted average paid.

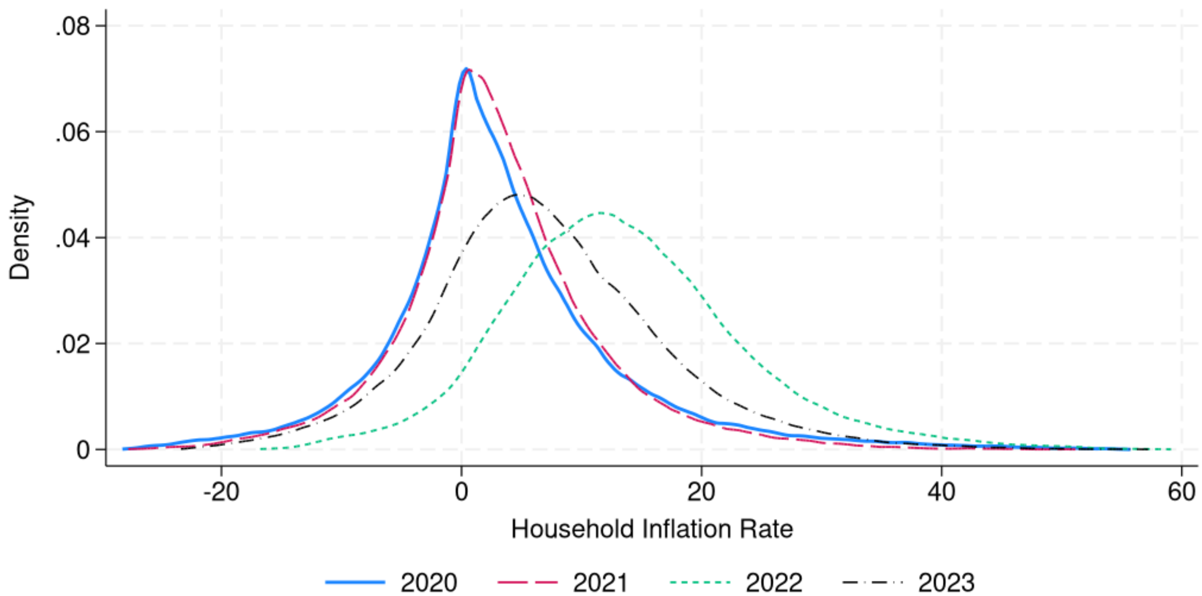
interpreting magnitudes and for constructing the shift-share instrument (which uses national CPI category inflation as the common shock component). I provide additional discussion of the Laspeyres index compared to alternative indices and the implications in Appendix B.

I focus on household-level inflation rates (specifically, year-over-year percent changes in a household's price index) rather than absolute price levels. The central question in my analysis is how changes in the cost of living influence household shopping behavior. It is these changes, rather than the absolute cost of the basket, that plausibly trigger adjustments in shopping effort and consumption patterns. While a month-over-month change could in principle capture the most immediate price pressures faced by households, it poses two problems. First, prices for many goods display strong seasonal patterns (e.g., produce, holiday items, school supplies), which can create spurious inflation signals unrelated to underlying cost-of-living changes. A year-over-year comparison naturally differences out this seasonal component by comparing the same month across years.

Year-over-year inflation also aligns with how inflation is commonly reported and discussed in public discourse. Headline CPI statistics are typically presented as 12-month changes, and households likely form expectations and make spending decisions based on these familiar benchmarks rather than on short-run monthly volatility. Using a 12-month horizon therefore better captures the inflation concept that is salient to consumers and relevant for behavioral adjustment.

Figure 4 shows the distribution of household-level inflation rates in 2019, 2021, and 2022. Several features stand out. First, consistent with other analyses of household inflation heterogeneity, I find substantial cross-sectional dispersion even within the same time period, despite all households facing the same aggregate economic conditions and shopping in overlapping sets of stores. Second, the distribution shifts rightward between 2019 and 2022, consistent with the aggregate increase in food inflation over this period. This confirms that the household-level measures capture the broad inflationary episode, not just idiosyncratic variation in individual shopping patterns.

Figure 4. Distribution of Household Inflation Rates by Year.



Third, and most importantly for my analysis, the distribution widens considerably in 2022 relative to 2019, indicating that inflation did not affect all households equally; instead, the variance in inflation experiences grew as aggregate inflation rose.

The widening of the household inflation distribution in 2022 has direct implications for the behavioral responses I study. In models of consumer search, the return to shopping effort depends on price dispersion: when prices vary more across products or stores, the potential gains from comparison shopping increase. If inflation raises the variance of price changes across categories, it increases the heterogeneity in effective prices faced by households, raising the returns to search. The widening distribution I document is consistent with this mechanism. It implies that the inflationary period was not simply a uniform scaling up of all prices but rather a period of heightened relative price volatility, which should amplify search incentives.

Additionally, the heterogeneity in household inflation rates provides the identifying variation for my empirical strategy. Because households differ in their 2019 consumption patterns, they experience systematically different inflation rates when national category-level prices evolve heterogeneously. A household with a high share of eggs in 2019 faces much higher inflation in 2022 (when egg prices surge due to avian flu) than a household with a low egg share, even if both live in the same city and shop at the same stores. This differential exposure, driven by predetermined consumption shares interacted with national price shocks, forms the basis of my shift-share instrument. The widening of the distribution in high-inflation periods amplifies this identifying variation, strengthening the first stage and improving the precision of my estimates.

The cross-sectional heterogeneity in inflation experiences may also help explain heterogeneity in behavioral responses. If different households face different magnitudes of inflation shocks, and if the marginal returns to search vary across the inflation distribution (for instance, due to nonlinearities or fixed costs of adjusting shopping routines), then households should respond differently in ways that correlate with their inflation exposure. In Section [X], I explore this heterogeneity explicitly by estimating treatment effects across income groups, household size, and initial shopping intensity. The pattern in Figure 4 motivates this analysis, as the fact that households experience widely varying inflation rates suggests that their behavioral adjustments may also vary, both in magnitude and in the margins along which they adjust.

I also report the distribution of household inflation over time in Figure A1, which shows the same increase in the variance of the distribution during times of higher inflation.

A natural concern with any household-level inflation measure constructed from data like the Consumer Panel is whether it accurately captures price movements as measured by official statistics. To validate my approach, I aggregate individual household inflation rates to construct a sample-wide price index and compare it to the published CPI for food at home. Specifically, I construct an expenditure-weighted Laspeyres index across all households in my sample, using the same methodology as for individual households but pooling across the entire panel. This aggregation mimics the structure of the official CPI, which weights category price changes by expenditure shares.

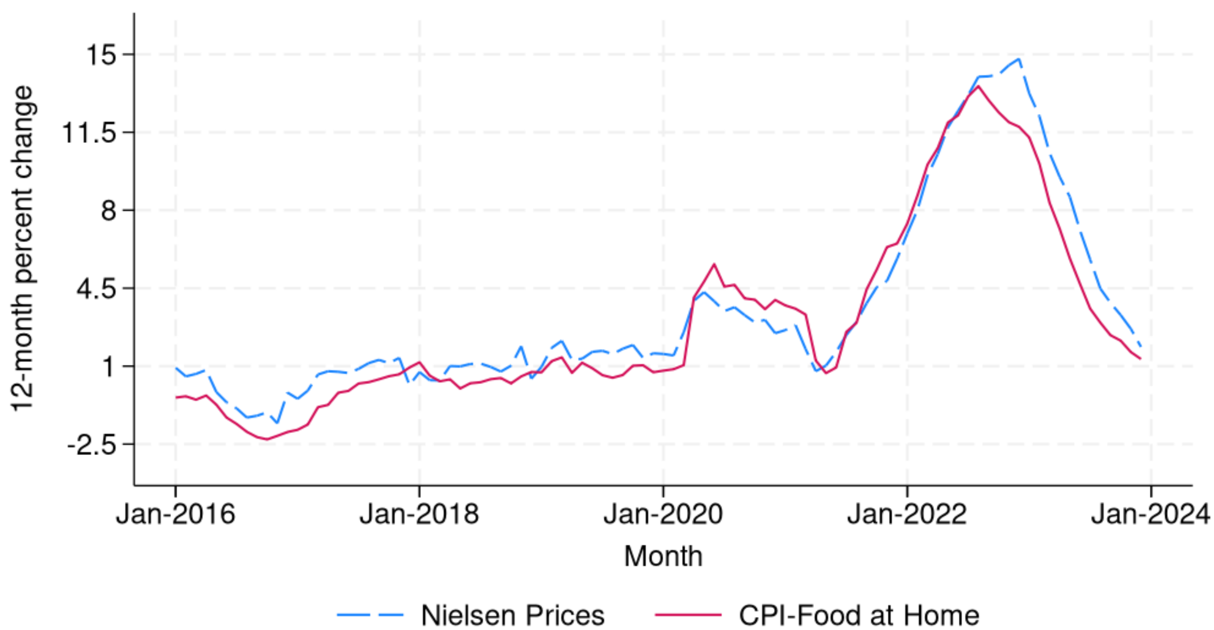
Figure 5 plots both series over the 2019 to 2023 period. The constructed measure tracks headline CPI for food at home closely throughout the sample period, both in levels and in the timing of inflections. Importantly, the constructed index captures the sharp acceleration in food inflation during 2021 and 2022, the peak in mid-2022, and the subsequent deceleration in 2023. This close correspondence provides confidence that the household-level inflation rates I construct from NielsenIQ data are measuring the same underlying price dynamics as the official statistics.

The similarity between my measure and the CPI is reassuring on several dimensions. First, it suggests that the UPC matching procedure and quality controls I impose (requiring at least 5 matched UPCs per household-month) are not introducing systematic bias. If my sample were disproportionately capturing price changes for a narrow set of products or missing key categories, the aggregate index would likely diverge from the CPI. Second, it validates the use of national CPI category-level shocks in the shift-share instrument. Because my constructed household inflation measure aggregates to the same dynamics as the official

CPI, this suggests that national category-level shocks I use as shifts are indeed the relevant price movements faced by households in the sample. Third, it implies that any remaining measurement error in household-level inflation is likely classical (uncorrelated with true inflation), supporting the interpretation that IV estimation corrects for attenuation bias.

Minor deviations between the two series are expected and do not threaten the validity of the approach. The NielsenIQ Consumer Panel covers a subset of food categories and retail channels, while the CPI Food-at-home potentially includes a broader set of outlets and uses a different quality adjustment methodology. Additionally, to compute the various CPI measures, the BLS uses a fixed basket updated periodically, while my household-level measures allow the comparison set of UPCs to vary as products enter and exit the market. These differences allow me to construct household-specific inflation rates that reflect actual shopping behavior while still tracking aggregate price movements closely enough to ensure comparability with national statistics.

Figure 5. Aggregate Inflation Series.



5 Empirical Methodology

To measure the impacts of household inflation on shopping and consumption outcomes, I estimate the following regression model using OLS:

$$y_{it} = \alpha_1 + \beta_1 \pi_{it} + \mu_{i1} + \delta_{t1} + \varepsilon_{it1} \tag{12}$$

where i indexes households and t indexes time periods. The variable y_{it} is an outcome, such as the number of shopping trips a household takes or the share of items bought on sale in a given shopping trip, and π_{it} is the household-level realized inflation rate computed from its actual prices paid. The variables μ_i and δ_t represent household and time fixed effects, respectively, so β_1 is identified by changes in an individual household's inflation rate over time. I include household fixed effects to control for unobserved time-invariant characteristics at the household-level. The time effects control for common shocks to all households at a given month. I also include combined statistical area (CSA) x month fixed effects to control for regional, time-specific shocks, so that I compare individuals in the same market at the same time. I cluster standard errors at the household level.

My empirical strategy requires tracking households across multiple years to observe how basket costs evolve over time. Specifically, I use pre-pandemic consumption shares from 2019 to capture inflation exposure while avoiding endogenous feedback from behavioral responses to price changes. The Consumer Panel is available from 2004 to 2023, and I use monthly data from 2019-2023 in my analysis.

5.1 Instrumenting for Household Inflation

As I describe in Section 3, when estimating the impact of household inflation on shopping and consumption outcomes, the problem resembles a system of linear simultaneous equations where the OLS estimate of β_1 will yield inconsistent and biased results. The ideal experiment would provide variation that moves individual inflation rates without directly affecting shopping behavior through channels other than prices. To address this endogeneity concern, I construct an instrumental variable that provides plausibly exogenous variation in individual exposure to inflation.

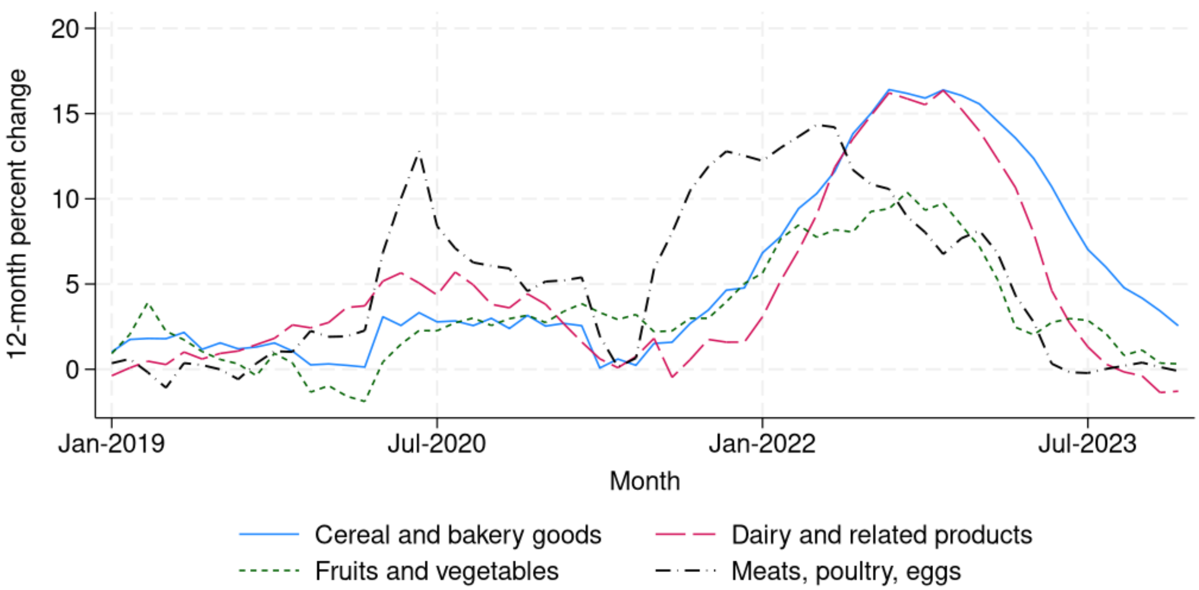
I employ a Bartik-style shift-share instrument (SSIV) that combines temporal variation in inflation at the CPI-stratum level with cross-sectional variation in predetermined consumption patterns:

$$\text{SSIV}_{it} = \sum_{k=1}^{N} s_{ik}^{2019} \times \pi_{kt}^{CPI} \tag{13}$$

where i indexes households, k indexes product groups (matched to NielsenIQ Consumer Panel categories), and t indexes time periods. This instrument predicts individual household inflation rates in period t as a weighted average of national product-group inflation rates, using pre-pandemic (2019) household consumption shares as weights. The CPI category inflation rates are a set of shifts common to all units, as they are at the national level, while the consumption categories are the sets of exposure shares that vary across units.

The insight motivating my identification strategy is the substantial heterogeneity in price dynamics across product categories, both in timing and magnitude of price changes. Figure 6 illustrates this variation by showing price growth for major nondurable categories during the recent inflationary period. This heterogeneity means that households with different consumption patterns experience systematically different inflation even when subject to the same aggregate economic conditions. A household whose 2019 basket emphasized eggs, for example, was exposed to much higher inflation than a neighbor who consumed more bread and fewer poultry products, due to variation in price dynamics of these categories.

Figure 6. Variation in Inflation Across CPI Components.



Source: Bureau of Labor Statistics.

Given that my research design emphasizes differential individual exposure to national inflation shocks across CPI subcategories, I argue that identification follows from the exogeneity of the shifts, rather than the shares (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). Household-level category-level expenditure growth rate shifts reflect changes to both behavior and to prices; to isolate price variation, I can form an instrument where I choose a set of common shifts to replace the household ones. These shifts are meant to be predictive of the household-level inflation rates in each category while only capturing variation in prices at the national level, clean of household behavior and its subsequent effect on their prices paid. The instrument $SSIV_{it} = \sum_k s_{ik,2019}\,\pi_{kt}$ uses fixed, pre-period shares $s_{ik,2019}$ as exposure weights on national shocks π_{kt}. These shares need not be exogenous in levels, and households with different base-year shares may systematically differ in observables or persistent tastes. Household fixed effects absorb such level differences. Identification instead comes from the time series variation in national stratum price shocks interacted with predetermined shares. The maintained restriction is that, conditional on fixed effects and controls, innovations in π_{kt} are orthogonal to the time-varying component of unobservables affecting outcomes, so that these shocks influence behavior only through realized household inflation. In other words, the exclusion restriction requires that national CPI category-level price shocks do not directly affect individual shopping behavior except through their impact on the prices that households actually pay.

It is important to distinguish two distinct forms of endogeneity in this setting. First, households with different 2019 consumption shares may systematically differ in observables (income, location, preferences) or persistent unobservables (brand loyalty, store access). This type of endogeneity is permitted and absorbed by household fixed effects. The instrument does not require that households with high egg shares are otherwise identical to households with high bread shares; it only requires that the differential inflation experienced by these two types of households (due to divergent price paths for eggs versus bread) is not itself correlated with differential trends in their shopping behavior for reasons unrelated to inflation.

The exclusion restriction rules out a different type of endogeneity: that national price shocks systematically target specific households at specific times in ways that affect outcomes beyond the price channel. For example, if egg prices rise nationally because of avian flu, the exclusion restriction requires that this shock affects households with high egg shares only because they face higher inflation, not because the shock independently alters their shopping patterns through some other mechanism (such as heightened food safety concerns that lead them to shop more carefully regardless of prices).

5.2 Potential identification threats

Several channels could potentially violate the exclusion restriction. First, national price shocks may be accompanied by supply-side disruptions that directly affect shopping behavior. For instance, if a category experiences both price increases and stockouts, households might make additional trips to find substitute products, confounding the price effect with an availability effect. To address this concern, I include CSA by month fixed effects, which absorb any regional supply shocks, retailer-level promotional strategies, or local economic conditions that correlate with national price movements. Identification therefore comes from comparing households within the same local market at the same time who differ in their exposure to national price shocks based on predetermined consumption patterns.

Another identification concern is that national inflation shocks could correlate with changes in retailer behavior that directly facilitate or hinder search. For example, if categories with high inflation see retailers respond by increasing promotional intensity or reducing product variety, the instrument might capture these supply responses rather than pure demand-side search adjustments. Two features of the research design mitigate this concern. First, the outcomes I examine (trips, store diversification, coupon usage) are primarily demand-driven and reflect household effort rather than retailer supply. Second, to the extent that retailer responses are common within a market, the CSA by month fixed effects absorb them. Any remaining variation comes from differential household exposure to national shocks, which is determined by pre-period shares that retailers cannot observe or target.

One might worry that national CPI shocks are themselves endogenous to aggregate shopping behavior. If many households simultaneously reduce consumption of a category, this could both lower demand and raise relative prices (through general equilibrium effects), creating reverse causality. However, this concern is mitigated by the fact that the CPI category-level shocks I use are driven by national and often global factors (commodity prices, supply chain disruptions, weather shocks) that are plausibly orthogonal to the shopping behavior of any individual household in the NielsenIQ Consumer Panel. The share of national aggregate consumption represented by NielsenIQ panelists is small, so their behavior cannot meaningfully influence national price dynamics.

The exclusion restriction implicitly assumes that category-level inflation does not vary systematically for households exerting higher or lower shopping effort within a category. If households that search more intensively within a category also systematically face lower inflation in that category (because they find better deals), and if this search intensity is itself correlated with the instrument, then the IV estimates could be biased. However, this concern is attenuated by the structure of the instrument. Because the shifts are at the national CPI level, they do not reflect individual household search effort. A household with a high egg share in 2019 faces the same national egg price shock as one with a low egg share, regardless of how much either household searches. The instrument generates variation in inflation exposure through the interaction of national shocks with predetermined shares, not through endogenous within-category search.

Again, the identifying assumption is that, conditional on household fixed effects, time fixed effects, and CSA by month fixed effects, the variation in household inflation induced by the shift-share instrument is orthogonal to unobserved determinants of shopping behavior. Households with different 2019 shares may differ in levels (absorbed by fixed effects), but the timing of national price shocks interacted with those shares provides exogenous variation in inflation exposure that affects behavior only through the price channel.

6 Effects of Household Inflation on Shopping Outcomes

The first stage is given by

$$\pi_{it} = \gamma_0 + \gamma_1 SSIV_{it} + \mathbf{X}_{it}\Gamma_1 + \theta_{i1} + \lambda_{t1} + \nu_{it1} \tag{14}$$

where γ_1 represents the percentage point increase in actual household inflation rates associated with a 1pp increase in SSIV-predicted household inflation rates.

The reduced form equation directly regresses outcomes on the instrument:

$$y_{it} = \alpha_2 + \beta_2 SSIV_{it} + \mathbf{X}_{it}'\Pi_2 + \mu_{i2} + \delta_{t2} + \varepsilon_{it2} \tag{15}$$

where β_2 represents the change in shopping or consumption behavior associated with a 1 percentage point increase in predicted household inflation rates. The reduced form is informative on its own, as it captures the total effect of inflation exposure (as predicted by pre-period shares and national shocks) on outcomes, without requiring assumptions about the exclusion restriction. A positive and significant β_2 indicates that households exposed to higher predicted inflation adjust their behavior accordingly.

The 2SLS coefficient of interest is $\beta_{IV} = \frac{\beta_2}{\gamma_1}$, which scales the reduced form effect by the strength of the first stage. This coefficient recovers the causal effect of a 1 percentage point increase in realized household inflation on outcomes, purged of simultaneity bias and measurement error. Intuitively, β_{IV} answers the question, when a household experiences higher inflation (as induced by differential exposure to national price shocks), how does its shopping behavior respond?

All specifications include household fixed effects, which absorb time-invariant differences across households. Time fixed effects control for common aggregate shocks that affect all households in a given month. I also include combined statistical area (CSA) by month fixed effects to account for regional, time-specific variation in economic conditions, ensuring that identification comes from comparing households within the same local market at the same time. Standard errors are clustered at the household level to allow for arbitrary correlation in the error term within households over time.

6.1 Baseline Results

Table 1 reports the effects of household inflation on shopping frequency and diversification. OLS estimates are small and negative, consistent with simultaneity: households that increase search effort tend to pay lower prices, mechanically lowering measured paid inflation and biasing OLS downward. By contrast, the reduced form and 2SLS estimates are positive and substantially larger, consistent with attenuation from measurement error and simultaneity bias. The shift-share instrument addresses simultaneity by isolating variation in paid inflation that stems from national price shocks interacted with pre-period expenditure shares, recovering the causal effect of inflation exposure on search behavior.

In the IV specification, a 1pp increase in paid inflation raises monthly shopping trips by 1.95 off a baseline mean of 12.15, about a 16 percent increase. The same 1pp increase expands the number of distinct retailers visited by 0.59 on a mean of 5.41 ($\approx$ 11%), and raises trips per store by 0.076, approximately 4 percent of the mean 1.91. Together, these patterns indicate that households respond to inflation by both intensifying visits on existing margins and diversifying across more retailers.

Table 1. Effects of Higher Inflation: Shopping Frequency and Retailer Diversification.

Dependent variable:	Trips	Distinct Retailers	Trips Per Store
	(1)	(2)	(3)
Panel A: OLS			
$\pi_{i,t}$	-0.005***	-0.002***	0.000
	(0.001)	(0.000)	(0.000)
Panel B: Reduced Form			
Predicted π_{it}	0.459***	0.139***	0.018***
	(0.007)	(0.003)	(0.002)
Panel C: IV (2SLS)			
$\pi_{i,t}$	1.951***	0.591***	0.076***
	(0.133)	(0.041)	(0.010)
First-stage F stat.	228.71	228.71	228.71
Base Dep. Var. Mean	12.15	5.41	1.91
Observations	734,132	734,132	734,132

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

Let β_{RF} denote the reduced-form effect of the shift–share instrument $SSIV_{it}$ on an outcome and γ_1 the first-stage effect on paid inflation. The 2SLS coefficient $\beta_{IV} = \beta_{RF}/\gamma_1$ captures the causal effect of a 1pp increase in household paid inflation induced by national CPI category shocks interacting with pre-period household expenditure shares (the standard shift–share exposure). Thus β_{IV} is a LATE for compliers whose paid inflation loads more heavily on the instrument due to their 2019 consumption mix. The sign flip from OLS to IV is consistent with (i) simultaneity (greater search lowers paid inflation) biasing OLS downward, and (ii) measurement error in household inflation attenuating OLS toward zero. The large and precise first-stage F-statistics (all greater than 225) indicate strong instruments across outcomes.

Table 2 shows similar responses in bargain-hunting within trips. The IV estimates imply that a 1pp rise in paid inflation increases the share of trips with a coupon by 0.812 percentage points relative to a mean of 12.8 percent (a 6.3 percent increase, approximately), and the share of trips with at least one on-sale item by 0.914pp relative to a mean of 30.4 percent (approximate increase of 3.0 percent). These results point to within-trip re-optimization toward promotions—consistent with models where higher expected gains from search during inflation raise optimal search intensity.

Table 2. Effects of Higher Inflation: Propensity for Bargain-Hunting During Shopping Trips.

Dependent variable:	Used Coupon	Bought an On-Sale Item
	(1)	(2)
Panel A: OLS		
$\pi_{i,t}$	-0.046***	-0.021***
	(0.002)	(0.003)
Panel B: Reduced Form		
Predicted π_{it}	0.195***	0.220***
	(0.016)	(0.021)
Panel C: IV (2SLS)		
$\pi_{i,t}$	0.812***	0.914***
	(0.086)	(0.107)
First-stage F stat.	243.56	243.56
Base Dep. Var. Mean	12.80	30.38
Observations	748,378	748,378

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

Table 3 documents substitution toward lower-cost retailers. A 1pp increase in paid inflation raises monthly trips to discount, dollar, and warehouse retailers by 0.421, 0.122, and 0.150, respectively. Relative to means of 2.25, 0.67, and 0.67, these are increases of roughly 18.7 percent, 8.2 percent, and 22.4 percent. This reallocation toward value-oriented outlets aligns with the diversification margins above. The shift toward these kinds of retailers is consistent with rational search behavior under inflation. Discount grocers, dollar stores, and warehouse clubs typically offer lower everyday prices and emphasize private-label goods, making them natural destinations for households seeking to reduce expenditures without dramatically cutting quantities. Warehouse formats additionally allow bulk purchasing, which can yield per-unit savings when combined with adequate storage capacity and liquidity to afford larger upfront outlays. The fact that all three value formats see significant increases suggests households are not simply substituting to the single cheapest option but rather broadening their retail portfolio to exploit the comparative advantages of different channels (everyday low prices at discount grocers, small package sizes at dollar stores, bulk discounts at warehouses).

Table 3. Effects of Higher Inflation: Trips to Lower-Cost Retailers.

Dependent variable:	Discount Store	Dollar Store	Warehouse Store
	(1)	(2)	(3)
Panel A: OLS			
$\pi_{i,t}$	-0.003***	-0.000	-0.001***
	(0.000)	(0.000)	(0.000)
Panel B: Reduced Form			
Predicted π_{it}	0.100***	0.029***	0.036***
	(0.002)	(0.001)	(0.001)
Panel C: IV (2SLS)			
$\pi_{i,t}$	0.421***	0.122***	0.150***
	(0.030)	(0.010)	(0.011)
First-stage F-stat.	231.60	231.60	231.60
Base Dep. Var. Mean	2.25	0.67	0.67
Observations	732,866	732,866	732,866

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

A useful benchmark for interpreting the magnitude of format substitution is to ask: of the 1.95 additional trips induced by a 1 percentage point increase in paid inflation, how many are directed to value formats? The point estimates imply roughly 0.69 trips to discount, dollar, and warehouse formats combined, representing approximately 35 percent of the total increase in trip frequency. This is substantially higher than the baseline share of trips to value formats in the sample, confirming that the marginal trips are disproportionately (and strategically) allocated to channels where price savings are likely to be greatest. The remaining trips include both additional visits to conventional grocers (potentially timed to promotions) and exploratory trips to broaden the household's retailer set, both of which are consistent with heightened search intensity.

Table 4 shows compositional changes in the basket. The IV coefficient on the generic (store-brand) share of expenditure is +0.29pp relative to a mean of 17.47 percent (a 1.7 percent increase), consistent with quality downgrading to preserve real consumption.

Table 4. Effects of Higher Inflation: Basket Composition.

Dependent variable:	Generic Items (% of Expenditure)
	(1)
Panel A: OLS	
$\pi_{i,t}$	0.004**
	(0.001)
Panel B: Reduced Form	
Predicted π_{it}	0.069***
	(0.010)
Panel C: IV (2SLS)	
$\pi_{i,t}$	0.290***
	(0.046)
First-stage F-stat.	239.62
Base Dep. Var. Mean	17.47
Observations	749,588

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

Table 5 indicates more stocking-up. A 1pp increase in paid inflation raises items per trip by 1.00 relative to a mean of 12.82 (a 7.8 percent increase), and increases items per pack by 0.021 on a mean of 1.57 (a 1.3 percent increase). These responses suggest households better time purchases and tilt toward multi-packs or larger bundle sizes when inflation is high.

Table 5. Effects of Higher Inflation: Stocking Up and Bulk-Buying.

Dependent variable:	Items Per Trip	Items Per Pack
	(1)	(2)
Panel A: OLS		
$\pi_{i,t}$	0.000	-0.000
	(0.001)	(0.000)
Panel B: Reduced Form		
Predicted π_{it}	0.236***	0.005***
	(0.008)	(0.001)
Panel C: IV (2SLS)		
$\pi_{i,t}$	1.002***	0.021***
	(0.074)	(0.004)
First-stage F stat.	228.16	228.16
Base Dep. Var. Mean	12.82	1.57
Observations	740,651	740,651

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

Finally, Table 6 shows that the number of distinct product groups purchased rises by 15.7 on a mean of 64.3 (an approximate 24.4 percent increase). Along with more retailers and value formats, this points to broader search across categories, consistent with active substitution to maintain purchasing power.

Table 6. Effects of Higher Inflation: Basket Variety.

Dependent variable:	Number of Product Groups
	(1)
Panel A: OLS	
$\pi_{i,t}$	-0.021***
	(0.004)
Panel B: Reduced Form	
Predicted π_{it}	3.759***
	(0.034)
Panel C: IV (2SLS)	
$\pi_{i,t}$	15.693***
	(1.039)
First-stage F stat.	237.29
Base Dep. Var. Mean	64.31
Observations	744,226

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

At first glance, the simultaneous rise in trips per month and items per trip may appear contradictory. If households are simply making more trips, one might expect baskets to shrink as the same total quantity is spread across more shopping occasions. The data show the opposite. This is consistent with strategic trip timing: households appear to concentrate larger stock-up trips around promotional events (weekend circulars, coupon availability) while interspersing smaller top-up visits. What I measure as "items per trip" therefore reflects an average that tilts upward because the promotion-timed trips become substantially larger. The fact that the number of product categories purchased also rises sharply (24.4 percent relative to the mean) reinforces this interpretation. Households are not merely buying more of the same items but are broadening the scope of comparison shopping and taking advantage of deals across a wider range of goods.

Several features of the data help distinguish the search mechanism from competing explanations. First, the strong responses on effort-intensive margins (trips, store diversification, and value-format substitution) are difficult to attribute purely to supply-side changes in promotion availability. If retailers simply offered more deals during high-inflation periods, I might observe higher coupon/sale usage, but the expansion in trips and cross-store shopping points to active household behavior. Second, the combination of more trips with larger baskets and broader category coverage argues against a stockout or inventory-driven story. If supply disruptions were forcing households to make extra trips to find unavailable goods, I would expect smaller baskets and fewer categories per trip; instead, I observe the reverse. Third, the shift toward value formats is not mechanically implied by the instrument. National CPI shocks do not directly affect the relative attractiveness of discount versus conventional grocers, suggesting genuine substitution in response to inflation exposure. Finally, the modest magnitude of generic share increases (compared to the large trip and format effects) indicates that households exhaust search-based strategies before resorting to substantial quality downgrading.

While the baseline specifications estimate linear semi-elasticities, there are theoretical reasons to suspect that responses may be convex in inflation exposure. As inflation rises, price dispersion typically increases, making the returns to search grow more than proportionally. Empirically, this would manifest as larger per-unit responses at higher levels of paid inflation. Similarly, the time dimension matters: search adjustments may exhibit modest persistence as households experiment with new shopping habits and update their information sets about store prices and sale schedules.

6.2 Time Costs

To obtain an estimate of the cost of these behaviors, I draw on the American Time Use Survey (ATUS). As one part of the survey, the ATUS asks respondents to provide a diary of activities for one day, which capture both the type and duration of activities including time spent grocery shopping. From this, I compute the average grocery shopping trip recorded by 41,793 respondents on their diary days from 2003 to 2024 to last approximately 43 minutes. Using my IV estimate, a 1 percentage point increase in household inflation raises the number of shopping trips by about 1.95 trips per month. Given an average trip length of 43 minutes, this translates into roughly 1.4 additional hours of shopping per month for each percentage point of inflation. Valuing that time at a nominal wage of $36 per hour, the monthly cost is about $50 per 1pp of inflation. Sustaining 1pp of inflation for a full year imposes a time cost worth approximately $600 per household. For larger inflation episodes, the burden scales proportionally: 5pp sustained inflation implies a cost of about $3,000 per year, while 8pp inflation implies a cost near $4,800.[4]

7 Conclusion

In this paper, I examine how household-level inflation exposure drives specific behavioral responses during the post-pandemic inflationary episode. Using granular household shopping data, I focus on bargain-hunting behaviors that households reported as primary coping mechanisms, such seeking sales and using coupons, as well as behaviors that are associated with intense product search, such as taking more trips and shopping at a greater variety of retailers. Taken together, these measures capture key behavioral margins that households can adjust when facing inflationary pressures without necessarily needing to fundamentally alter the composition of their basket. With these, rather than immediately substituting away from preferred goods, households can first intensify their search efforts and deal-seeking behaviors to maintain original consumption while still minimizing expenditure increases. This behavioral response that I find is consistent with models where households face search costs but can benefit from increased search when the payoffs to finding lower prices rise with inflation.

By documenting these causal responses during a unique and highly salient inflationary period, I provide evidence of households experience and react to realized inflation, and my results demonstrate that consumer behavior is more dynamic and costly than solely

[4] My time-cost calculations assume (i) linear effects in $\Delta\pi$ and (ii) the average trip length is constant at 43 minutes. I do not compute time costs for visiting distinct retailers, as the coefficient captures diversification of store choice but does not map directly to additional time without further assumptions about within-trip multi-store visits.

substituting to cheaper, lower quality goods. These behavioral involve real resource costs, particularly in terms of time spent searching across retailers and monitoring prices, and survey evidence indicates that many households are indeed partaking in these shopping strategies. Future work should explore how long such adaptations can be sustained, and how ongoing shocks, such as tariffs or supply disruptions, may interact with these behaviors. Taken together, these findings show the consequences of inflation, in that it affects households not only via prices paid but also in the hidden costs of search, effort, and time.

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A Additional Tables and Figures

Figure A1. Distribution of Household-Level Inflation Rates.

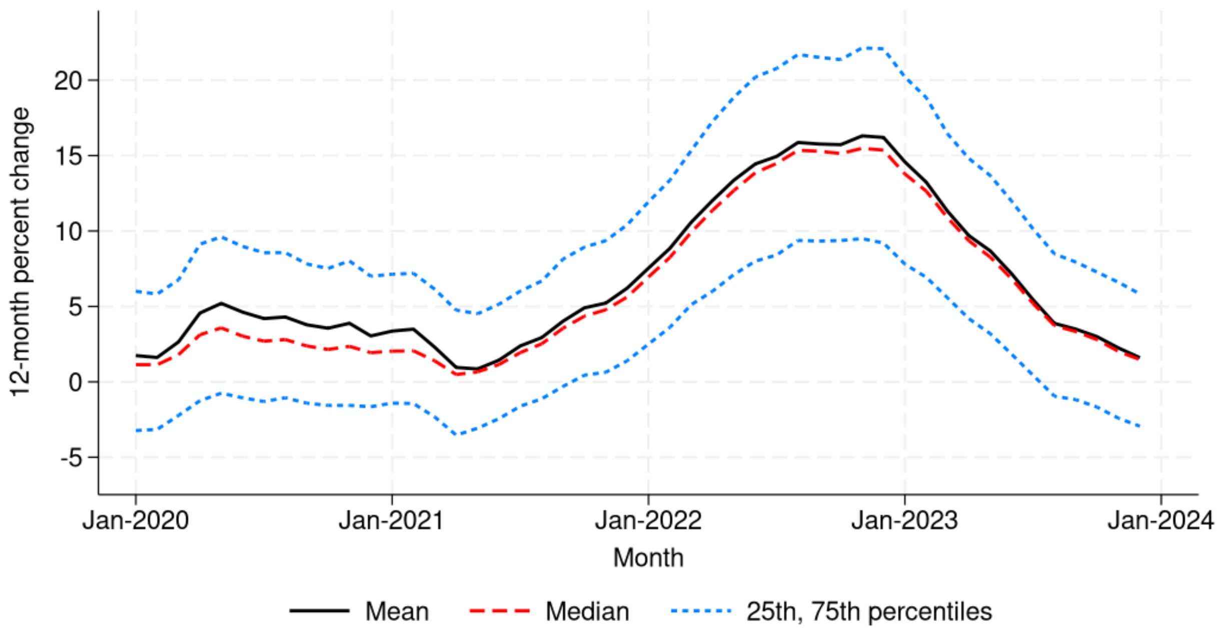


Table A1. Monthly Bargain-Hunting Item Totals.

Dependent variable:	Coupon Items	Sale Items
	(1)	(2)
Panel A: OLS		
$\pi_{i,t}$	-0.029***	-0.033***
	(0.003)	(0.001)
Panel B: Reduced Form		
Predicted π_{it}	0.957***	0.255***
	(0.022)	(0.009)
Panel C: IV (2SLS)		
$\pi_{i,t}$	3.980***	1.060***
	(0.275)	(0.079)
First-stage F-stat.	240.23	240.23
Base Dep. Var. Mean	21.25	5.36
Observations	733,431	733,431

Notes: Each column reports estimates from separate regressions of shopping outcomes on the specified independent variable. The sample includes data from 2021 to 2023. Fixed effects include individual, month, and county x month. Standard errors are clustered at the household level and are in parentheses. * - significant at 10%, ** - significant at 5%, *** - significant at 1%.

B Household Inflation Rates - Alternative Indices

Below I explain alternative ways to construct household inflation rates. The Laspeyres index, which I use in my baseline analysis, is defined as follows:

$$P^L_{i,t,t+12} = \frac{\sum_j p_{ij,t+12}\, q_{ij,t}}{\sum_j p_{ij,t}\, q_{ij,t}}. \tag{16}$$

By construction, $P^L_{i,t,t} = 1$, so the associated inflation rate is simply the percent change in the index:

$$\pi^L_{i,t,t+12} = (P^L_{i,t,t+12} - 1) * 100. \tag{17}$$

The Paasche index instead reweights prices by the final-period quantities:

$$P^P_{i,t,t+12} = \frac{\sum_j p_{ij,t+12}\, q_{ij,t+12}}{\sum_j p_{ij,t}\, q_{ij,t+12}}, \tag{18}$$

with the corresponding inflation rate

$$\pi^P_{i,t,t+12} = (P^P_{i,t,t+12} - 1) * 100. \tag{19}$$

Finally, the Fisher index takes the geometric mean of the Laspeyres and Paasche indexes:

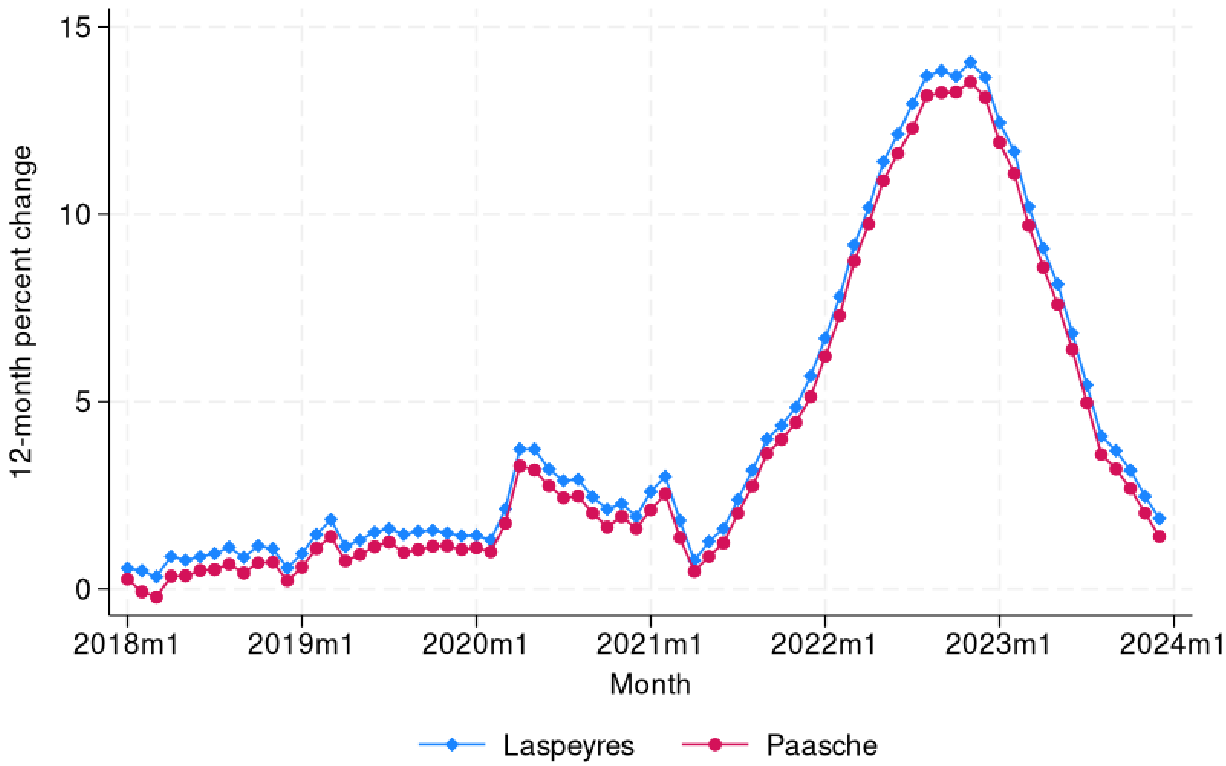
$$P^F_{i,t,t+12} = \sqrt{P^L_{i,t,t+12}\, P^P_{i,t,t+12}}, \tag{20}$$

and its inflation rate is

$$\pi^F_{i,t,t+12} = (P^F_{i,t,t+12} - 1) * 100. \tag{21}$$

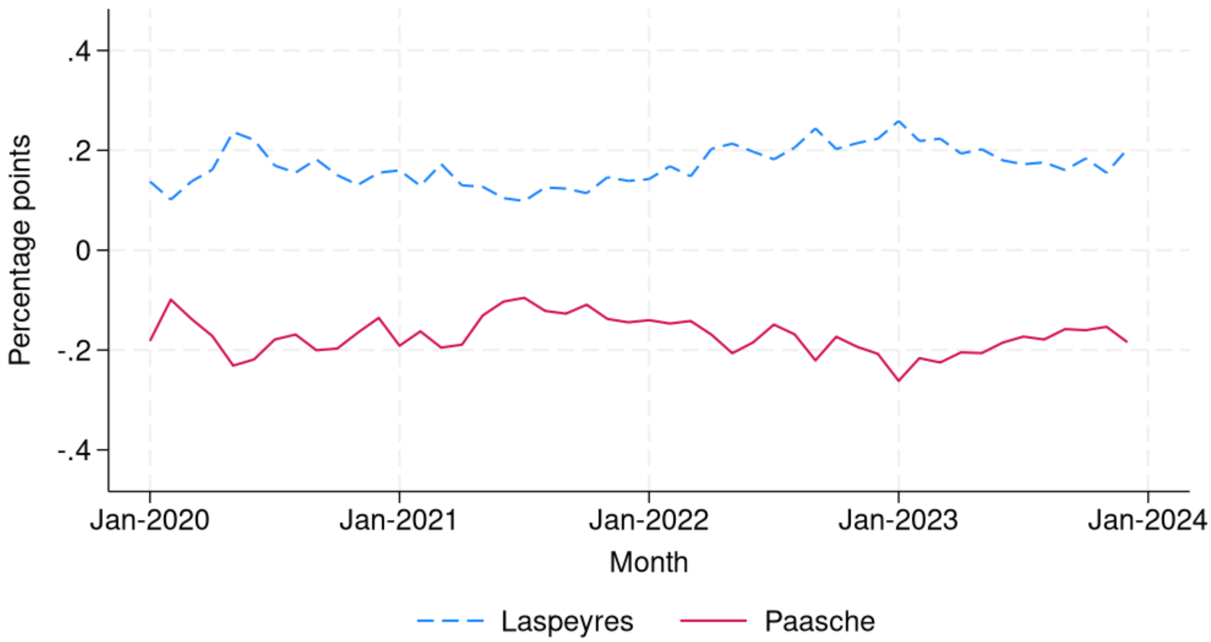
When relative prices shift, more expensive goods receive greater weight in the Laspeyres index (since weights come from the pre-substitution period), while cheaper goods receive greater weight in the Paasche index (since weights come from the post-substitution period). As a result, the Laspeyres measure is typically higher than the Paasche. Figure A2 show this is the case for the household inflation rates I calculate in equations 17 and 19. This is because the Laspeyres index holds the initial-period basket fixed, implicitly assuming that households continue to purchase the same bundle of goods even after relative prices change. Because consumers typically substitute away from items that become more expensive, this assumption tends to overstate the true cost of living. In contrast, the Paasche index holds the final-period basket fixed, weighting price changes by the quantities actually consumed after substitution has taken place. This produces the opposite bias: it tends to understate the true cost of living, because it assumes households could have purchased the cheaper, substitution-

Figure A2. Aggregate Household-Level Inflation Rates.



adjusted bundle from the start. The Fisher index, defined as the geometric mean of the two, balances these opposing biases. It is often considered the best approximation to a true cost-of-living index, and the gap between the Laspeyres and the Fisher provides a measure of the substitution bias inherent in the Laspeyres. I present a time series of the mean difference from the Fisher index for both the Laspeyres and the Paasche in Figure A3.

Figure A3. Mean Differences in Laspeyres and Paasche Indices from Fisher Index.



C Shift-Share Instrument Construction

In this section, I describe the data and sources used to construct the shift-share instrument introduced in Section 5.1.

C.1 CPI Component Data for Instrument Construction

To construct the SSIV, I use disaggregated CPI data published monthly by the Bureau of Labor Statistics (BLS). The CPI measures aggregate price changes using a representative basket methodology that combines household expenditure surveys with extensive price collection from retail establishments.

I match NielsenIQ's product group classifications to corresponding CPI subcategories to construct household-specific inflation exposure measures. Table A2 shows the mapping. This matching allows me to combine the detailed household consumption patterns from NielsenIQ with the national price trend data from CPI to predict individual household inflation rates using the shift-share approach I describe in Section 5.1.

CPI Series	NielsenIQ Product Group	CPI Series	NielsenIQ Product Group
Apparel	Hosiery, socks; Shoe Care; Sunglasses	Hair, dental, shaving, and miscellaneous personal care products	Personal soap and bath additives; Deodorant; Feminine hygiene; Grooming aids; Hair care; Men's toiletries; Oral hygiene; Sanitary protection; Shaving needs; Skin care preparations
Appliances	Kitchen gadgets	Household furnishings and supplies	Batteries and flashlights; Electronics, records, tapes; Housewares, appliances; Light bulbs, electric goods
Baby food and formula	Baby food; Disposable diapers; Baby needs	Housekeeping supplies	Household supplies
Bacon, breakfast sausage, and related products	Breakfast food	Ice cream and related products	Ice cream, novelties
Beer, ale, and other malt beverages at home	Beer	Juices and nonalcoholic drinks	Juice, drinks - canned, bottled
Recreational books	Books and magazines	Laundry equipment	Laundry supplies
Bread	Bread and baked goods; Dough products	Meats	Packaged meats - deli; Fresh meat
Butter and margarine	Butter and margarine	Medicinal drugs	Cough and cold remedies; Diet aids; Medications/remedies/health aids; Vitamins
Candy and chewing gum	Candy; Gum	Medical equipment and supplies	First aid
Shelf stable fish and seafood	Seafood - canned	Milk	Packaged milk and modifiers; Milk
Canned fruits and vegetables	Fruit - canned; Jams, jellies, spreads; Vegetables - canned	Olives, pickles, relishes	Pickles, olives, and relish
Carbonated drinks	Carbonated beverages; Soft drinks - non-carbonated	Other beverage materials including tea	Tea
Cereals and cereal products	Cereal	Other condiments	Table syrups, molasses
Cheese and related products	Cheese	Household paper products	Paper products; Greeting cards/party needs/novelties
Household cleaning products	Detergents; Fresheners and deodorizers; Household cleaners	Rice, pasta, cornmeal	Pasta
Coffee	Coffee	Pet food and treats	Pet food; Pet care
Cookies	Cookies	Prepared salads	Dressings/salads/prepared foods - deli
Nonelectric cookware and tableware	Baking supplies; Canning, freezing supplies; Cookware; Glassware, tableware	Other processed fruits and vegetables including dried	Fruit - dried; Vegetables and grains - dried; Desserts/fruits/toppings - frozen; Vegetables - frozen
Cosmetics, perfume, bath, nail preparations and implements	Cosmetics; Fragrances - women	Spices, seasonings, condiments, sauces	Spices, seasoning, extracts
Crackers, bread, and cracker products	Crackers	Sauces and gravies	Condiments, gravies, and sauces
Dairy and related products	Cottage cheese, sour cream, toppings; Yogurt	Sewing machines, fabric and supplies	Sewing notions
Salad dressing	Salad dressings, mayo, toppings	Snacks	Nuts; Snacks; Pizza, snacks, hor d'oeurves - frozen; Snacks, spreads, dips - dairy
Eggs	Eggs	Soups	Soup
Fats and oils	Shortening, oil	Distilled spirits at home	Liquor
Film and photographic supplies	Photographic supplies	Stationary, stationary supplies, gift wrap	Wrapping materials and bags; Stationary, school supplies
Flour and prepared mixes	Baking mixes; Flour; Yeast	Sugar and sugar substitutes	Sugar, sweeteners
Fresh fruit and vegetables	Fresh produce	Sugar and sweets	Pudding, desserts - dairy
Frozen and refrigerated bakery products, pies, tarts, turnovers	Baked goods - frozen	Tobacco and smoking products	Tobacco & accessories
Frozen and freeze dried prepared foods	Breakfast foods - frozen; Prepared foods - frozen	Tools, hardware and supplies	Automotive; Hardware, tools
Frozen fish and seafood	Unprepared meat/poultry/seafood - frozen	Toys	Toys & sporting goods
Frozen noncarbonated juices and drinks	Juices, drinks - frozen	Wine at home	Wine

Table A2

C.2 Product Category Shares

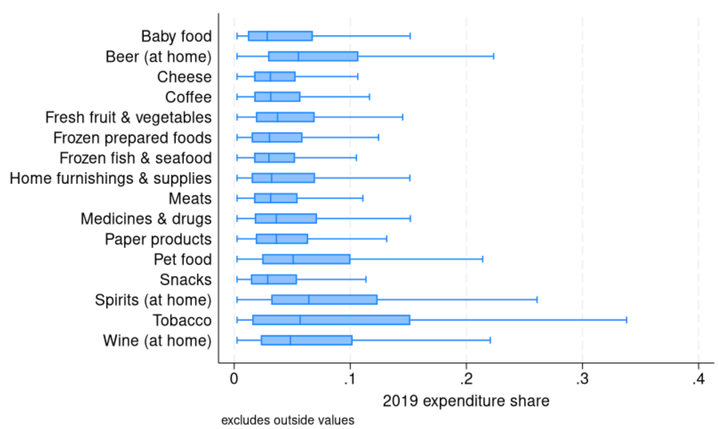
In addition to the national-level CPI data that I use for the shifts in the SSIV, I construct shares that measure each household's exposure to the inflation shocks in each category using the following formula:

$$s_{ij,t} \equiv \frac{\sum\limits_{k \in \mathcal{G}_j} p_{ik,t}\, q_{ik,t}}{E_{i,t}}, \qquad E_{i,t} \equiv \sum_{k \in \mathcal{K}_i(t)} p_{ik,t}\, q_{ik,t}.$$

where $\mathcal{G}_j$ the set of goods (barcodes) in category j, $\mathcal{K}_i(t)$ is the set of all goods purchased by household i in period t, $p_{ik,t}$ is the price paid by i for good k in t, and $q_{ik,t}$ is the quantity of good k purchased by i in t. I do this using expenditure data from 2019, as prices were already rising toward the end of 2020, and I want to ensure that the shares I use are effectively lagged to before the inflationary episode began. Though one might expect household consumption shares to have been altered considerably due to the pandemic in 2020, and therefore the 2019 shares would not be representative of households' true inflation exposure in 2021 and onward, I find that shares in the product groups that I use for my SSIV remain fairly stable since 2019.

In Figure A4 I plot a subset of the expenditure shares (by their corresponding CPI stratum category) that I use in my instrument in order to show some of the variation within and across groups.

Figure A4. Distribution of a subset of 2019 household expenditure shares.



C.3 Constructed instrument

I plot a time series of my instrument in Figure A5, as well as its distribution in Figure A6.

Figure A5. Predicted inflation instrument.

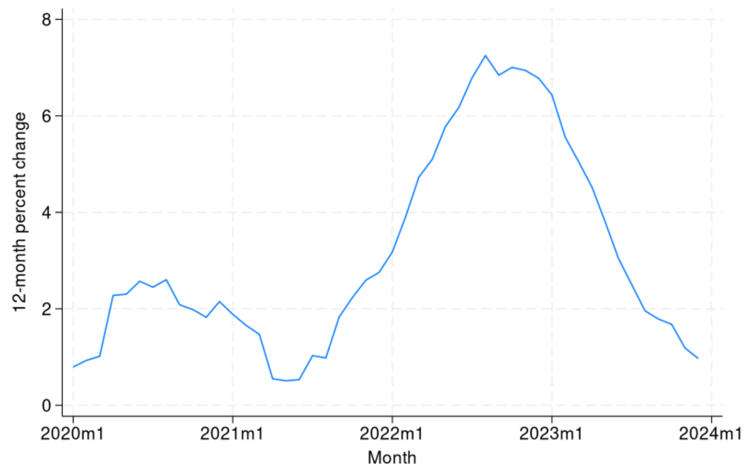


Figure A6. Predicted inflation instrument distribution.